

KIRTI KOUR

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Objective

Computer Science graduate with hands-on experience in full-stack web development and AI/ML systems. Co-author of IEEE-published research at iCoMET 2026 on computer vision and forensic AI. Built and deployed production-ready MERN applications and deep learning pipelines using React.js, Node.js, PyTorch, and Docker. STHP Merit Scholar actively seeking roles in full-stack development or AI/ML engineering.

Technical Skills

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, NLP, Computer Vision, LLM APIs (OpenAI, Anthropic/Claude), Prompt Engineering, AI Workflow Automation

Languages: Python, JavaScript, TypeScript, SQL, C++, Java

Frontend: React.js, Next.js, Vue.js, Tailwind CSS, Bootstrap, HTML, CSS, Figma

Backend: Node.js, Express.js, Django, FastAPI, Flask, REST APIs, GraphQL, JWT, OAuth

Databases: MongoDB, PostgreSQL, MySQL

DevOps & Tools: Docker, Git, GitHub, Linux, Vercel, Netlify, CI/CD, Postman

Experience

Full Stack Developer Intern

Elevo

2025

Remote / Pakistan

- Built JobTracker, a full-stack MERN job application platform with automated CV parsing, KPI dashboards, and JWT authentication delivered as part of production internship work.

Web Developer Intern

Sindh Empowered Oasis

2024

Sindh, Pakistan

- Developed and maintained 2 responsive WordPress websites with custom themes and plugins, collectively supporting 500+ weekly active users.

Research

Research Co-Author & Conference Presenter

IEEE ICOMET 2026

May 2026 📍

International Conference

- Co-authored and presented IEEE-published research on CNNs vs Vision Transformers for image manipulation detection, focusing on classification-localization trade-offs in forensic AI systems.
- Findings revealed that CNNs outperform Vision Transformers in localization tasks while ViTs demonstrate stronger classification performance – highlighting a fundamental architectural trade-off for forensic AI systems.

Projects

Solar Panel Segmentation via Self-Supervised Learning | *Flask, React.js, Docker, PyTorch, Linux*

FYP 📍

- Designed a multi-service architecture with Flask backend, React.js frontend, and Dockerized ML inference pipeline enabling real-time predictions via REST API.
- Benchmarked 5 self-supervised models; BYOL + U-Net achieved 93.38% IoU and 96.56% Dice Score as best-performing configuration.

HEALR – Medical Drug & Test Search Platform | *React.js, Node.js, Express.js, MongoDB*

2025 📍

- Built a full-stack MERN health platform with live search across 5,000+ medical entries, cart management, and appointment scheduling.
- Designed secure RESTful APIs with database pagination and filtering, improving client-side responsiveness by 25%.

JobTracker – Job Application Analytics Platform | *React.js, Node.js, Express.js, MongoDB, JWT*

2025 📍

- Built an automated CV parsing pipeline using NLP techniques tested across 10 resumes, auto-populating key profile fields and significantly reducing manual data entry time.
- Developed KPI tracking dashboard to monitor application status, trends, and outcomes with dark/light theme support and JWT-secured access.

Chest X-Ray Disease Analysis | *PyTorch, EfficientNet, React.js, Django*

2025 📍

- Preprocessed NIH ChestX-ray14 dataset for 8-class multi-label classification; handled class imbalance using Pandas and NumPy.
- Benchmarked 4 CNN architectures; EfficientNet-B0 achieved best performance at 74% Macro AUC, results delivered through React.js and Django reporting interface.

Education

Sukkur IBA University

Bachelor of Science in Computer Science

Sukkur, Pakistan

2022 – 2026

CGPA: 3.68/4.0

STHP Scholar (Merit-Based Fully Funded Scholarship)

Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, Software Engineering, Machine Learning, Enterprise Application Development, Information Security.

Certifications & Achievements

Certifications: Google AI Essentials – Coursera (2026) | Google Advanced Data Analytics – Coursera (2026) | Google Prompting Essentials – Coursera (2026) | Python and Django – University of Michigan (2024) | Web Development – IBM (2024)

Achievements: PM Laptop Scheme Awardee (2025) | STHP Merit Scholarship – Fully Funded BS Degree | Executive Member, CS Society, Sukkur IBA University | Hackathon Volunteer